

Bülent Köseoğlu

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EDUCATION

Yaşar University

B.S. in Computer Engineering (Senior Student)

Izmir, Turkey

Sept 2022 – July 2026 (Expected)

SELECTED PROJECTS

Automatic Digger/Tilling Machine - Autonomous Soil Processing System

May 2025

Technologies: Arduino Uno (ATmega328P), AVR C, WOKWI, Servo, HC-SR04 Ultrasonic Sensor, LCD

- Designed and developed a fully autonomous soil tilling mechanism based on Arduino Uno, aimed at minimizing manual labor in small-scale farming and hobby gardens.
- Implemented a closed-loop control system that monitors soil depth in real-time using an HC-SR04 ultrasonic sensor and dynamically adjusts the blade angle via a servomotor.
- Developed low-level firmware using AVR C, utilizing Timer1 overflow interrupts for high-precision depth measurement and external interrupts for system state management.
- Integrated safety protocols including obstacle detection via mechanical switches, triggering immediate system halts and multi-modal alerts (Buzzer/LED) upon detecting hard debris.
- Validated system performance in the WOKWI simulation environment through field-emulation scenarios, achieving a depth maintenance accuracy of ± 1 cm.

VR Boxing: Advanced Combat System & Infrastructure Architecture

April 2026

Technologies: Unity 6 (URP), C#, OpenXR, XR Interaction Toolkit, NavMesh

- Developed a physics-based VR boxing simulation optimized for Meta Quest and Valve Index platforms using Unity 6 and OpenXR.
- Engineered a "Punch Detector" system utilizing vector analysis and velocity tracking algorithms to automatically classify punch types such as Jab, Hook, and Uppercut.
- Integrated NavMesh and Finite State Machines (FSM) to create an advanced opponent AI capable of distance control, aggressive engagement, and evasion maneuvers.
- Utilized Continuous Collision Detection (CCD) and ConfigurableJoint components to ensure realistic physics interactions and prevent clipping during high-velocity impacts.
- Established a decoupled system architecture using Observer and Singleton patterns for synchronized real-time HUD updates, scoring, and spatial audio-visual effects.

Online Gym Management System - Digital Automation Platform

Jan 2026

Technologies: Web Technologies, OOP (Inheritance, Polymorphism, Encapsulation)

- Architected a web-based gym automation platform providing digital management for members, trainers, and administrators.
- Applied core OOP principles to build a scalable user management module where distinct roles are inherited from a centralized user structure.
- Developed a dynamic content infrastructure allowing trainers to manage personalized workout plans, exercise sets, and schedules, while enabling member progress tracking.
- Integrated an intelligent scheduling system for trainer availability and a direct messaging module to facilitate trainer-member communication.

Nessy Burger – Game Jam Project

Nov 2023

Technologies: Unity, C#

- Developed and published a mobile game on Itch.io within 48 hours for the Yaşar University Mobile Game Jam.

TECHNICAL SKILLS

Programming: C, C++, AVR C (Embedded), C#, OOP (Inheritance, Polymorphism, Encapsulation)

Frameworks & Tools: Unity 6, OpenXR, XR Interaction Toolkit, NavMesh, Arduino, WOKWI Simulation

Languages: English (B2 Level), Turkish (Native)